

Movie Review Sentiment Analysis

Naive Bayes Classification

Sakina Saifuddin

PS ID: 2512251237

EDS 6352 — Homework 1

1 Introduction and Method

This assignment is doing sentiment analysis of the positive and negative reviews from Kaggle dataset using naive bayes, it applies 4 different scenarios by using 4 different kinds of preprocessing techniques.

The dataset contained 40,000 reviews with two columns: `text` and `label`. Label 0 represented negative sentiment, and label 1 represented positive sentiment. No missing values were found in these columns.

The training sample contained 1,000 randomly selected reviews: 506 negative and 494 positive. The test sample contained 400 reviews, with 200 from each class. Training and test samples did not share any row indices.

Four preprocessing scenarios were evaluated using the same training and test samples:

1. Baseline: no lemmatization, stop-word removal, or negation handling.
2. Lemmatization only.
3. Lemmatization and stop-word removal.
4. Lemmatization, stop-word removal, and negation handling.

2 Evaluation Results

Table 1 reports the results from the executed notebook. Precision, recall, and F1 are reported separately for each class. Overall accuracy applies to all 400 test reviews and is repeated for both classes within each scenario.

Table 1: Evaluation results for the four preprocessing scenarios

Scenario	Class	Precision	Recall	F1	Accuracy
1	Negative	76.21%	86.50%	81.03%	79.75%
1	Positive	84.39%	73.00%	78.28%	79.75%
2	Negative	76.58%	85.00%	80.57%	79.50%
2	Positive	83.15%	74.00%	78.31%	79.50%
3	Negative	78.44%	85.50%	81.82%	81.00%
3	Positive	84.07%	76.50%	80.10%	81.00%
4	Negative	78.87%	84.00%	81.36%	80.75%
4	Positive	82.89%	77.50%	80.10%	80.75%

3 Answers to the Required Questions

3.1 Comparison of Scenarios and Recommendation

Scenario 3 that is Lemmatization + stop-word removal gave the highest total number of correct predictions across both the classes (+ve and -ve), and there's only 1 incorrectly predicted reviews difference between both scenario 3 and scenario 4. Lemmatization + stop-word removal gave the highest accuracy of 81.00%, handling negation correctly identified 2 more positive reviews but 3 fewer negative reviews compared to scenario 3, scenario 3 was itself enough to get the highest accuracy as compared to all the other 3 scenarios.

Without lemmatization, stop-word removal, and negation handling, there wasn't much difference in accuracy compared to the other scenarios, but statistical significance was not tested. Accuracy increased by 1.00 percentage point after adding all of these preprocessing techniques mentioned above, and by 1.25 percentage points with lemmatization + stop-word removal.

I'd choose scenario 3 Lemmatization + stop-word removal as a preprocessing combination in my dataset because it has the highest overall accuracy and negative-class f-1 score, although it does not have the highest precision and recall for both classes. So adding negation adds coding complexity without giving us the highest overall accuracy in this experiment, but just to compare I'd also compare it with handling negation to just for comparison.

3.2 Effect of Laplace Smoothing

By adding add-1 smoothing we make sure that every vocabulary word appeared atleast one extra time in each class in this case in out -ve and +ve classes. It makes sure that the one unseen word-class combination does not make the entire class score zero. This specifically happens because each likelihoods gets multiplied by each other and without adding 1 to the unseen word-class will make the entire result zero. So in Naive Bayes, smoothing avoids $\log(0)$ and in addition to that, denominator is also adjusted by adding the total vocab size to it, simply because we are adding 1 extra count to each word-class combination, so the count increases automatically, so we need to make sure that it reflects the right count.

3.3 Limitations and Error Analysis

In Naive Bayes, each word contributes seperately to the class score, so it does not fully undestand how the words together change the meaning of the review. This can be seen in some of the incorrectly predicted reviews in scenario 1.

For example, review 3088 was actually -ve but predicted as +ve. The reviewer says that the movie “leaves out two key things: action and thrills.” This means the movie is missing action and thrills, which is criticism. But the model does not understand the complete phrase like we do, so this could be one reason for the incorrect prediction.

Another example is review 7050, which was actually +ve but predicted as -ve. The reviewer questions why other people hate the movie and then says it is one of their favourites. Here, “hate” describes other people’s opinions, not the reviewer’s own opinion. The model may associate this word with negative reviews without understanding that difference.

These examples suggest that the model can struggle with context and different opinions within the same review. These are possible explanations, because we have not checked exactly how much each word contributed to these predictions. Preprocessing can help, but it does not make the model fully understand the meaning of the review.